Code No: 117FN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
MOBILE COMPUTING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What are the applications of mobile computing? [2]
b) List out the limitations of mobile and handheld devices. [3]
c) Define CDMA. [2]
d) What are the requirements for implementing Mobile IP? [3]
e) Define snooping TCP. [2]
f) Explain how API at mobile device sending queries and retrieving data from local database. [3]
g) Discuss about selective tunneling. [2]
h) Show Communication asymmetry in uplink and downlink in a mobile network. [3]
i) List the application of MANETS. [2]
j) Describe the protocols supported by Linux OS for mobile computing. [3]

PART-B

(50 Marks)

2. What are the subsystems used in GSM network architecture? Explain the functionality of each unit with GSM architecture. [10]

OR

3. With a neat diagram, explain the architectural layers of mobile computing. [10]

4. Write about various protocol specifications used in IEEE 802.11 physical layer. [10]

OR

5. Explain how SDMA is used to allocate separated space to users in wireless networks.[10]

6. What are the additional features of mobile TCP to handle fast retransmit/recovery and transmission/time-out freezing? Explain. [10]

OR

7. Write about distributed Hoarding (caching) of specific database in mobile devices with diagram. [10]

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8. Draw the diagram for mobile device pulling the data records from a server or set of distributed systems and also explain its bandwidth and thresholds. [10]

OR

9. What is broadcasting? Explain its architecture with applications. [10]

10. With neat sketch explain the architecture of WAP and its operational support. [10]

OR

11. How to pass messages using Dynamic Source Routing algorithm? Explain with example. [10]

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